

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY KAKINADA

Regulation: R23 | Semester: I | Branch: Common to All Branches

Basic Civil & Mechanical Engineering

UNIT - II

Manufacturing & Thermal Engineering

Topics Covered: Manufacturing Processes · Advanced Manufacturing · Thermal Engineering · IC Engines · Modern Vehicles

Prepared as per JNTUK R23 Syllabus

Includes: Theory · Diagrams · Formulae · Q&A · Quick Revision Sheet · References

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Section 1: Introduction to Manufacturing Processes

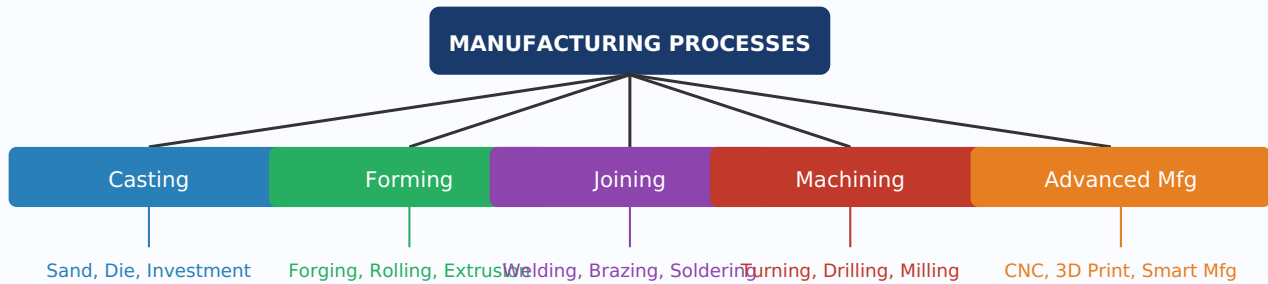
Foundation concepts | Classification | Importance

1.1 Definition of Manufacturing

Manufacturing is the process of converting raw materials into finished or semi-finished products through the application of mechanical, thermal, chemical, or other processes. The word originates from Latin *manu factus* meaning "made by hand." In modern context, it includes automated and computer-controlled processes.

1.2 Classification of Manufacturing Processes

FIG 1.1 - CLASSIFICATION OF MANUFACTURING PROCESSES (FLOWCHART)



1.3 Importance of Manufacturing

- Converts raw materials into useful products for daily life and industry.
- Drives economic growth, employment, and national productivity.
- Foundation of all engineering industries – automobile, aerospace, electronics, construction.
- Modern manufacturing integrates automation, AI, and IoT (Industry 4.0).

Key Concept: Manufacturing is broadly divided into **Primary processes** (casting, forming), **Secondary processes** (machining, finishing), and **Joining processes** (welding, fastening).

Section 2: Casting Process

Sand Casting | Principles | Diagram | Applications

2.1 Definition

Casting is a manufacturing process in which molten metal is poured into a mould cavity of the desired shape and allowed to solidify. After solidification, the solidified metal (called a *casting*) is removed from the mould.

2.2 Types of Casting

- **Sand Casting:** Most common; mould made from sand.
- **Die Casting:** Molten metal forced into a metal mould under pressure.
- **Investment Casting:** Wax pattern coated with ceramic slurry; used for precision parts.
- **Centrifugal Casting:** Metal poured into a rotating mould; used for pipes and cylinders.

2.3 Sand Casting - Working Principle (Step-by-Step)

Step 1: A **pattern** (replica of the final part) is placed in a moulding box (flask).

Step 2: Sand mixed with binder is packed around the pattern to form the **mould cavity**.

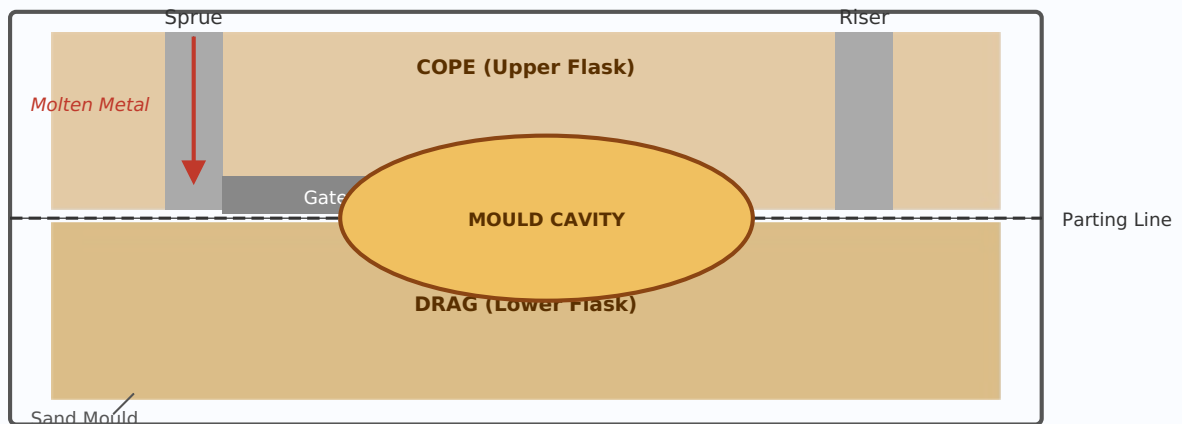
Step 3: The pattern is withdrawn, leaving the cavity. Gates and risers are cut.

Step 4: **Molten metal** is poured through the sprue/gate into the mould cavity.

Step 5: The metal **solidifies and cools** inside the mould.

Step 6: The mould is broken and the casting is **removed and cleaned** (fettling).

FIG 2.1 - SAND CASTING PROCESS DIAGRAM (CROSS-SECTION VIEW)



Casting - Key Formulae

Solidification Time (Chvorinov's Rule): $t = B \times (V/A)^2$

where t = solidification time, B = mould constant, V = volume of casting, A = surface area

Yield = (Weight of Casting / Weight of Metal Poured) $\times$ 100%

✓ Advantages

- Complex internal and external shapes possible
- Any metal/alloy can be cast
- Large components produced economically
- Low tooling cost for sand casting

✗ Disadvantages

- Surface finish is poor (sand casting)
- Dimensional accuracy limited
- Defects like porosity, shrinkage possible
- Labour intensive and time consuming

Applications:

Engine blocks, cylinder heads, pump casings, machine tool bodies, artistic sculptures, manhole covers, pipes, valves.

Section 3: Forming Processes

Forging | Rolling | Extrusion | Drawing

3.1 Definition

Forming processes involve shaping solid metal by applying mechanical forces (compression, tension, shear) without removing any material. The metal may be worked *hot* (above recrystallisation temperature) or *cold*.

3.2 Types of Forming

- **Forging:** Metal heated and shaped by compressive forces using dies and hammers.
- **Rolling:** Metal passed through rotating rolls to reduce thickness (plates, sheets, rails).
- **Extrusion:** Metal forced through a die orifice to produce long shapes of constant cross-section (rods, tubes).
- **Drawing:** Metal pulled through a die to produce wires and tubes of smaller diameter.
- **Sheet Metal Forming:** Bending, blanking, deep drawing of thin sheets.

3.3 Forging - Working Principle

Step 1: A metal billet or ingot is heated to forging temperature (hot forging: $0.6-0.9 \times$ melting point).

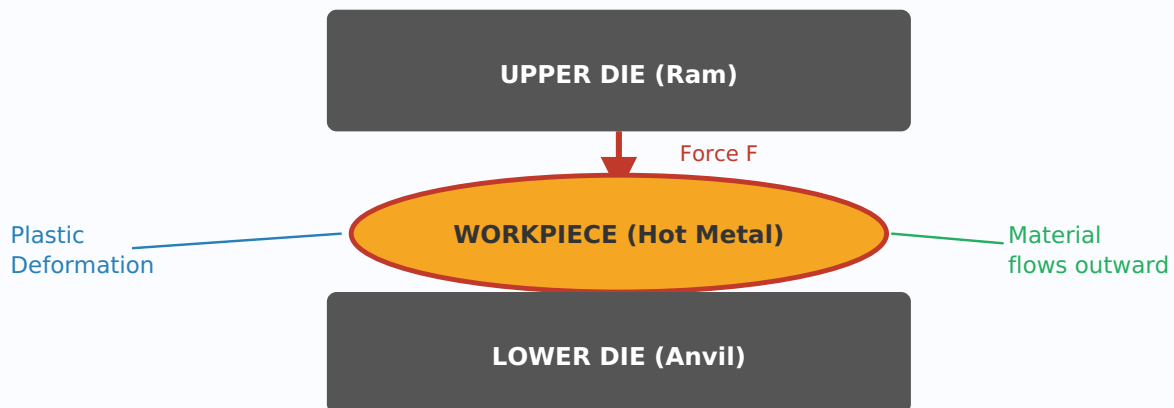
Step 2: The heated metal is placed between dies (upper and lower).

Step 3: A hammer or press applies compressive impact force.

Step 4: Metal deforms plastically and fills the die cavity.

Step 5: The forged part is removed, trimmed, and heat treated.

FIG 3.1 - FORGING PROCESS DIAGRAM



Forming - Key Formulae

True Strain: $\epsilon = \ln(A_0/A_f) = \ln(L_f/L_0)$

Rolling: Draft (d) = $t_0 - t_f$ | Reduction ratio $r = d/t_0$

Extrusion Ratio: $R = A_0/A_f$ (where A_0 = initial area, A_f = final area)

✓ Advantages

- Good mechanical properties (grain refinement)
- High strength-to-weight ratio
- Minimal material wastage
- Suitable for mass production

✗ Disadvantages

- High tooling and die cost
- Limited to ductile materials
- Size limitations in cold forming

Applications:

Crankshafts, connecting rods (forging); railway rails, I-beams (rolling); aluminum sections, window frames (extrusion); electrical wires (drawing).

Section 4: Joining Processes

Welding | Brazing | Soldering | Mechanical Fastening

4.1 Definition

Joining processes are used to permanently or temporarily join two or more components or materials together. They are essential in fabrication of structures, vehicles, pipelines, and machinery.

4.2 Types of Joining

- **Welding:** Fusion joining by melting base metals with or without filler. (e.g., Arc Welding, MIG, TIG, Spot Welding)
- **Brazing:** Filler metal with melting point above 450°C but below base metal melting point is used. Base metal does not melt.
- **Soldering:** Similar to brazing but filler melts below 450°C; used in electronics.
- **Mechanical Fastening:** Bolts, nuts, rivets, screws – temporary joints.
- **Adhesive Bonding:** Chemical adhesives used for lightweight/composite parts.

4.3 Arc Welding - Working Principle

Step 1: An electric arc is struck between the **electrode** and the **base metal**.

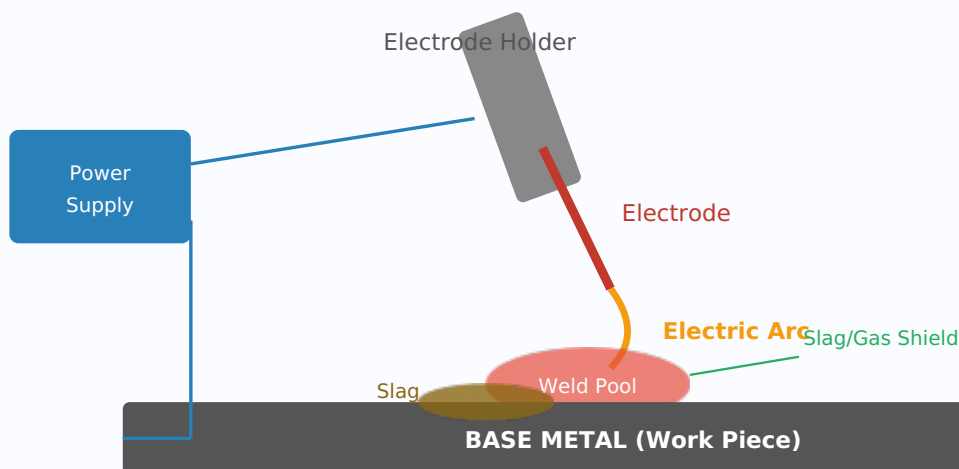
Step 2: The arc generates intense heat (~6000°C) that melts the electrode and base metal.

Step 3: A **weld pool** is formed by the molten metal.

Step 4: The flux covering the electrode forms **slag** that protects the weld from atmospheric contamination.

Step 5: The weld pool solidifies and the slag is chipped off to reveal the weld bead.

FIG 4.1 - ARC WELDING PROCESS DIAGRAM



Welding - Key Formulae

Heat Input: $H = (V \times I \times 60) / (S \times 1000) \text{ kJ/mm}$

where V = Voltage, I = Current (A), S = Travel speed (mm/min)

Efficiency of welding joint: $\eta = \text{Strength of joint} / \text{Strength of base metal}$

✓ Advantages

- Permanent and strong joints
- Suitable for complex structures
- Wide range of materials joinable
- Low-cost process

✗ Disadvantages

- Residual stresses and distortion
- Requires skilled operators
- Not easily disassembled
- HAZ affects base metal properties

Applications:

Bridges, ships, pipelines, automobiles, aerospace structures, pressure vessels, storage tanks.

Section 5: Machining Processes

Turning | Drilling | Milling | Lathe Machine

5.1 Definition

Machining is a material removal process where unwanted material is removed from the workpiece using a cutting tool to obtain the desired shape and dimensions. It is a subtractive manufacturing process.

5.2 Types of Machining

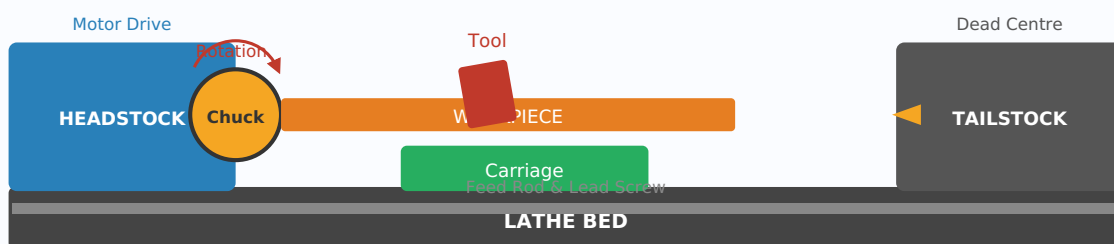
- **Turning:** Workpiece rotates; cutting tool moves linearly. Performed on a lathe.
- **Drilling:** Rotating drill bit creates cylindrical holes.
- **Milling:** Rotating multi-tooth cutter removes material from a fixed workpiece.
- **Grinding:** Abrasive wheel removes small amounts for finishing.
- **Shaping/Planing:** Reciprocating tool or workpiece for flat surfaces.

5.3 Lathe Machine - Construction & Working

A **lathe** is the mother of all machine tools. It holds the workpiece in a chuck and rotates it against a cutting tool to produce cylindrical shapes.

Main parts: Headstock, Tailstock, Carriage, Bed, Feed Rod, Lead Screw, Chuck, Tool Post.

FIG 5.1 - LATHE MACHINE DIAGRAM



Machining - Key Formulae

Cutting Speed: $V = \pi \times D \times N / 1000$ (m/min) [D=diameter mm, N=rpm]

Material Removal Rate (MRR) = $f \times d \times V$ (mm³/min) [f=feed, d=depth]

Machining Time: $T = L / (f \times N)$ (min) [L=length of cut]

✓ Advantages

- High dimensional accuracy
- Excellent surface finish
- Wide variety of materials

✗ Disadvantages

- Material is wasted as chips
- Time-consuming for complex parts
- Tool wear increases cost

Section 6: CNC Machines

Computer Numerical Control | Automation | Block Diagram

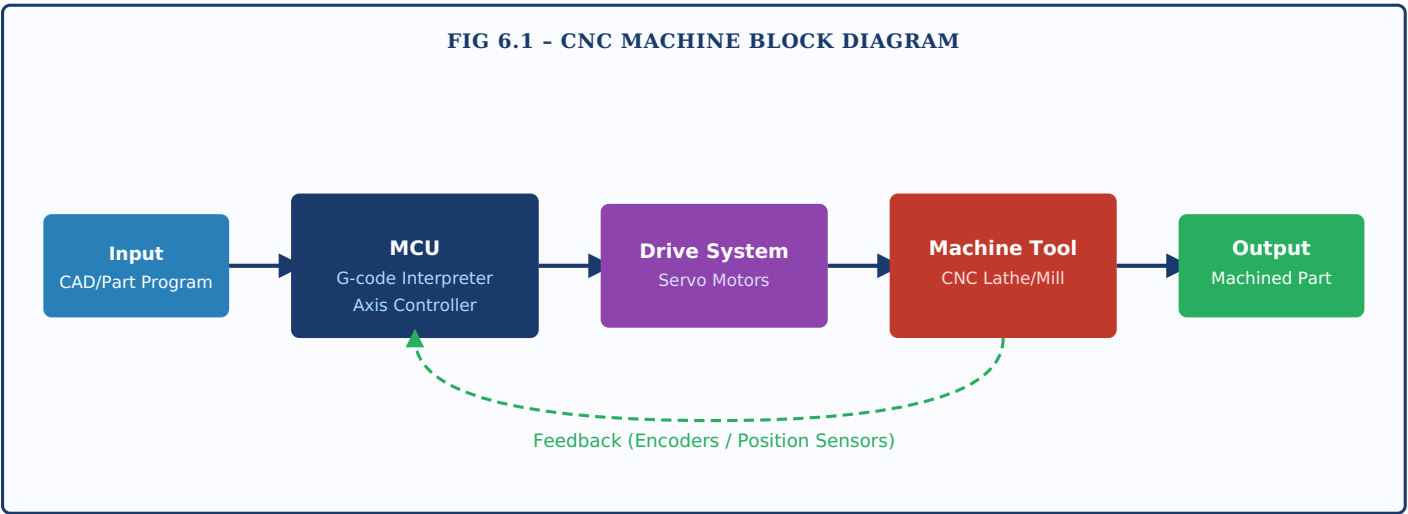
6.1 Definition

CNC (Computer Numerical Control) is a method of automating machine tools using a computer program. Instead of manual operation, the machine movements are controlled by pre-programmed sequences of commands (G-codes and M-codes).

6.2 Components of a CNC System

- **Input Device:** Keyboard, USB, or network to feed the part program.
- **Machine Control Unit (MCU):** The "brain" - interprets G-codes, generates axis commands.
- **Drive System:** Servo motors and stepper motors that move the machine axes.
- **Machine Tool:** Actual cutting/machining tool (lathe, milling, drilling).
- **Feedback System:** Encoders/sensors send actual position back to MCU for accuracy.

FIG 6.1 - CNC MACHINE BLOCK DIAGRAM



6.3 Advantages of CNC over Conventional Machines

Feature	CNC Machine	Conventional Machine
Accuracy	Very high (± 0.001 mm)	Moderate
Automation	Fully automated	Manual
Repeatability	Excellent	Operator-dependent
Skilled Labour	Programmer needed	Skilled machinist needed
Cost	High initial cost	Low initial cost
Production Rate	High	Low to medium

Applications:

Aerospace components, automotive parts, medical implants, mould and die making, electronics manufacturing.

Section 7: 3D Printing (Additive Manufacturing)

FDM | SLA | SLS | Layer-by-Layer Process

7.1 Definition

3D Printing or **Additive Manufacturing (AM)** is the process of creating three-dimensional objects by depositing material layer by layer, based on a digital 3D model (CAD file). Unlike subtractive machining, no material is wasted.

7.2 Types of 3D Printing

- **FDM (Fused Deposition Modelling):** Thermoplastic filament melted and extruded layer by layer. Most common desktop printer.
- **SLA (Stereolithography):** UV laser cures liquid photopolymer resin layer by layer. High resolution.
- **SLS (Selective Laser Sintering):** Laser fuses powdered material (nylon, metal). No support structure needed.
- **DMLS/SLM:** Metal powder sintered/melted for functional metal parts.

7.3 FDM Process - Step-by-Step

Step 1: Create a 3D model using CAD software (SolidWorks, AutoCAD, Fusion 360).

Step 2: Convert the CAD file to STL (Standard Tessellation Language) format.

Step 3: Slice the STL file into thin horizontal layers using slicing software (Cura, Simplify3D).

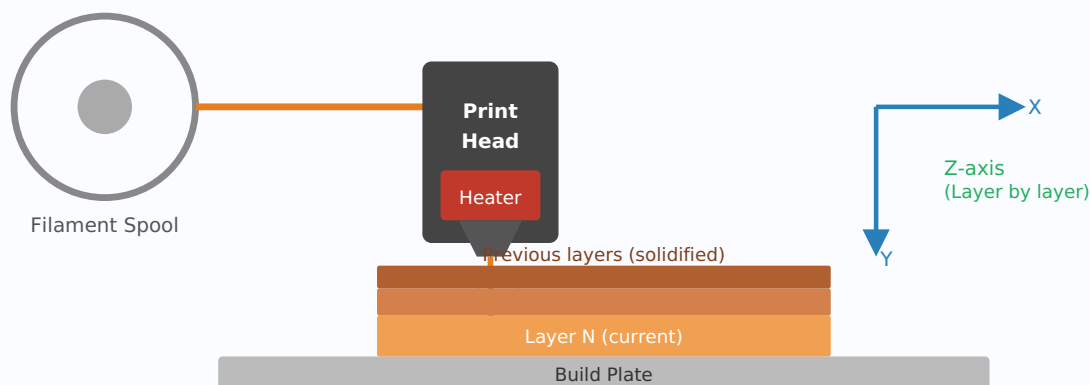
Step 4: The printer extrudes heated thermoplastic filament (PLA, ABS) through a nozzle.

Step 5: Material is deposited layer by layer on the build plate.

Step 6: Each layer fuses with the previous one as it cools and solidifies.

Step 7: Support structures (if any) are removed and post-processing is done.

FIG 7.1 - FDM 3D PRINTING PROCESS DIAGRAM



✓ Advantages

- Complex geometries without tooling
- Rapid prototyping – fast design validation
- Minimal material waste
- Customisation for every part

✗ Disadvantages

- Slow for large-volume production
- Limited material options (FDM)
- Anisotropic mechanical properties (layer lines)
- Post-processing required

Applications:

Medical implants and prosthetics, aerospace lightweight structures, architectural models, dental crowns, electronics housings, jewellery, educational models.

Section 8: Smart Manufacturing

Industry 4.0 | IoT | AI | Cyber-Physical Systems

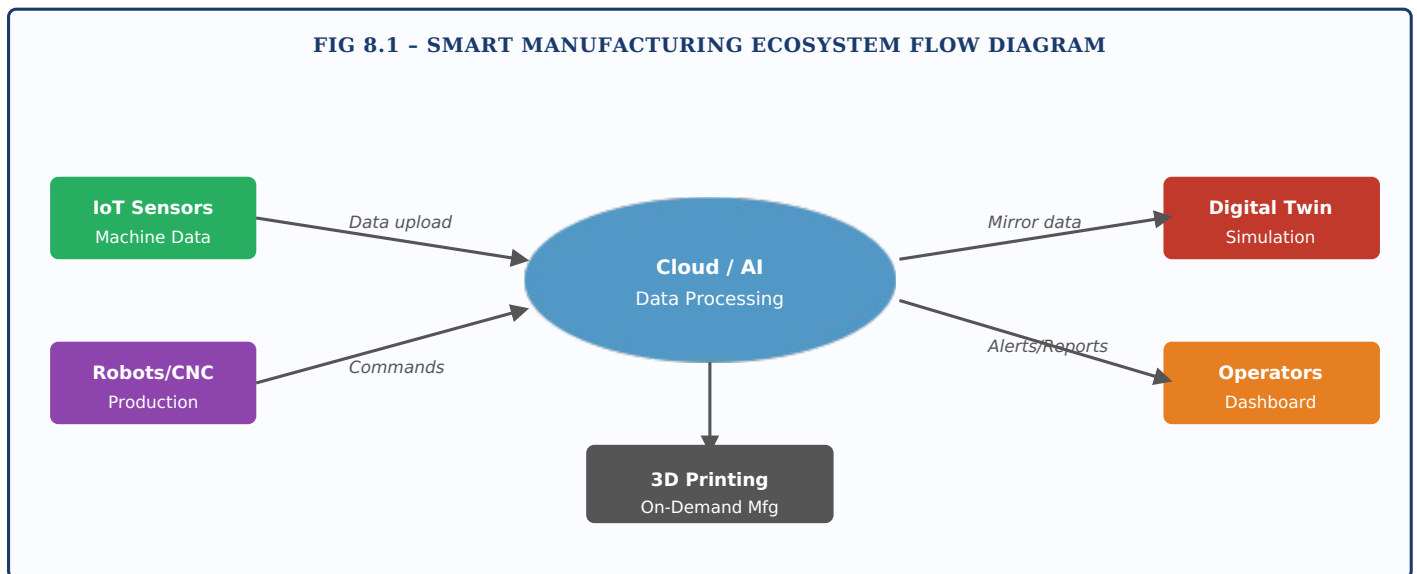
8.1 Definition

Smart Manufacturing (Industry 4.0) is an advanced manufacturing approach that integrates **Artificial Intelligence (AI)**, **Internet of Things (IoT)**, **Big Data**, **Cloud Computing**, and **Cyber-Physical Systems (CPS)** to create intelligent, self-optimising production systems.

8.2 Key Technologies in Smart Manufacturing

- **IoT (Internet of Things):** Sensors on machines collect real-time data; connected to the internet.
- **AI & Machine Learning:** Predictive maintenance, quality control, process optimisation.
- **Digital Twin:** A virtual replica of physical machines for simulation and monitoring.
- **Robotics & Cobots:** Automated robots working alongside humans (collaborative robots).
- **Cloud Manufacturing:** Manufacturing data stored and processed in the cloud.
- **Additive Manufacturing:** On-demand, customised 3D printing integrated with digital workflows.

FIG 8.1 - SMART MANUFACTURING ECOSYSTEM FLOW DIAGRAM



Applications:

Automobile assembly plants, semiconductor fabrication, pharmaceutical manufacturing, food processing, aerospace component production, predictive maintenance systems.

Section 9: Thermal Engineering Basics & Boilers

Thermodynamic Laws | Steam Boilers | Cochran Boiler

9.1 Introduction to Thermal Engineering

Thermal Engineering is the branch of engineering dealing with the conversion of thermal energy (heat) into mechanical work and vice versa. It applies the principles of thermodynamics, heat transfer, and fluid mechanics.

Laws of Thermodynamics

Zeroth Law: If $A = C$ and $B = C$ in temperature, then $A = B$ (Thermal Equilibrium)

First Law: $Q = \Delta U + W$ (Heat added = Change in internal energy + Work done)

Second Law: Heat flows naturally from hot to cold; no 100% efficient engine.

Efficiency of Heat Engine: $\eta = 1 - (T_L / T_H) = W_{net} / Q_H$

9.2 Boiler - Definition

A **Boiler** is a closed pressure vessel in which water is heated beyond its boiling point to generate steam under pressure. The steam generated is used for power generation, industrial processes, and space heating.

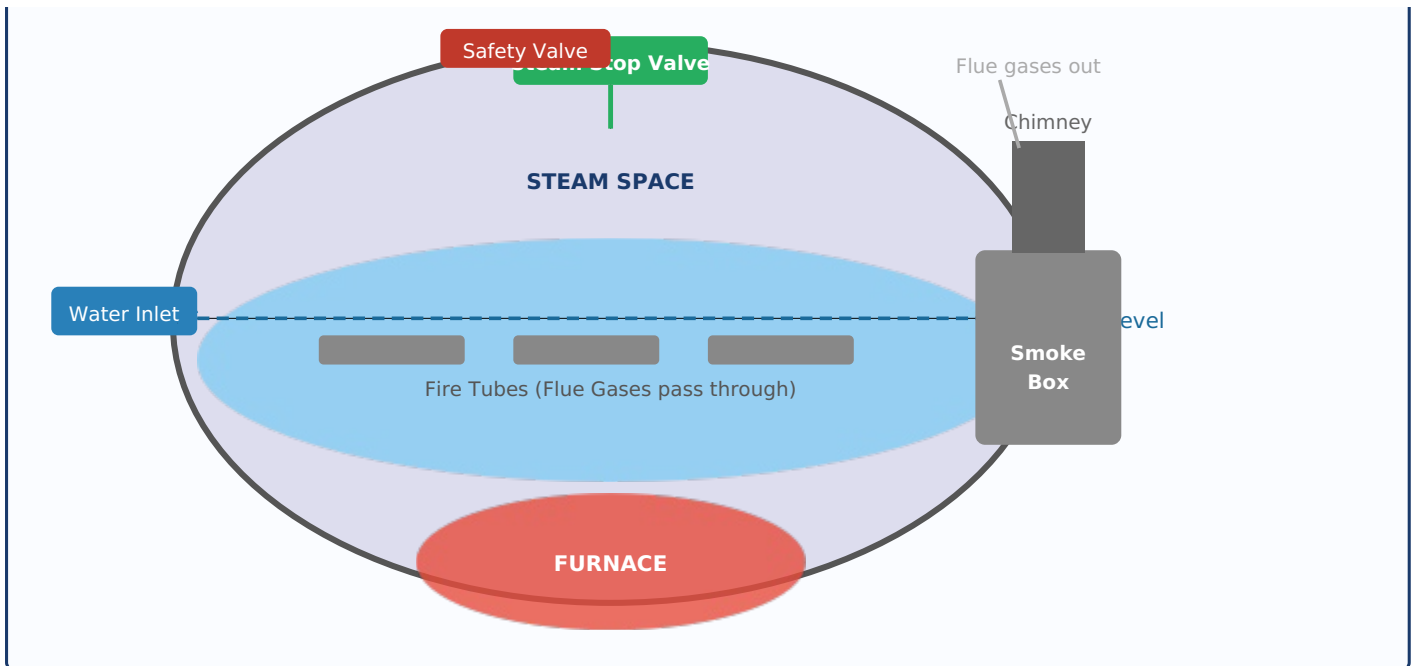
9.3 Classification of Boilers

- **Fire Tube Boiler:** Hot flue gases pass through tubes surrounded by water (Cochran, Lancashire, Cornish boiler).
- **Water Tube Boiler:** Water flows through tubes surrounded by hot gases (Babcock & Wilcox boiler). High pressure & capacity.
- **Based on Position:** Vertical (Cochran) and Horizontal (Lancashire).
- **Based on Circulation:** Natural circulation and Forced circulation.

9.4 Cochran Boiler - Working Principle

- Step 1:** Fuel is burned in the **furnace** at the bottom of the boiler.
- Step 2:** Hot flue gases rise and pass through a **combustion chamber**.
- Step 3:** Gases flow through multiple horizontal **fire tubes** submerged in water.
- Step 4:** Heat transfers from gases to water surrounding the tubes, generating steam.
- Step 5:** Steam collects in the **steam space** above the water level.
- Step 6:** Steam is withdrawn through the **steam stop valve** for use.

FIG 9.1 - COCHRAN (FIRE TUBE) BOILER DIAGRAM



Boiler - Key Formulae

Equivalent Evaporation: $EE = \frac{m_s(h_s - h_{fw})}{(2257) \text{ kg steam/kg fuel}}$

Boiler Efficiency: $\eta_b = \frac{m_s \times (h_s - h_{fw})}{(m_f \times CV)} \times 100\%$

where m_s = steam generated, h_s = enthalpy of steam, h_{fw} = enthalpy of feed water

m_f = fuel consumed, CV = calorific value of fuel

Section 10: Otto Cycle & Diesel Cycle

Ideal Cycles for SI and CI Engines | P-V Diagrams

10.1 Otto Cycle (Constant Volume Cycle)

The **Otto Cycle** is the ideal thermodynamic cycle for Spark Ignition (SI/petrol) engines. It consists of **four reversible processes**:

- **Process 1→2**: Isentropic (adiabatic) compression of air-fuel mixture.
- **Process 2→3**: Constant Volume heat addition (combustion at TDC).
- **Process 3→4**: Isentropic (adiabatic) expansion (power stroke).
- **Process 4→1**: Constant Volume heat rejection (exhaust valve opens).

Otto Cycle - Efficiency Formula

Thermal Efficiency: $\eta_{\text{Otto}} = 1 - (1 / r^{(\gamma-1)})$

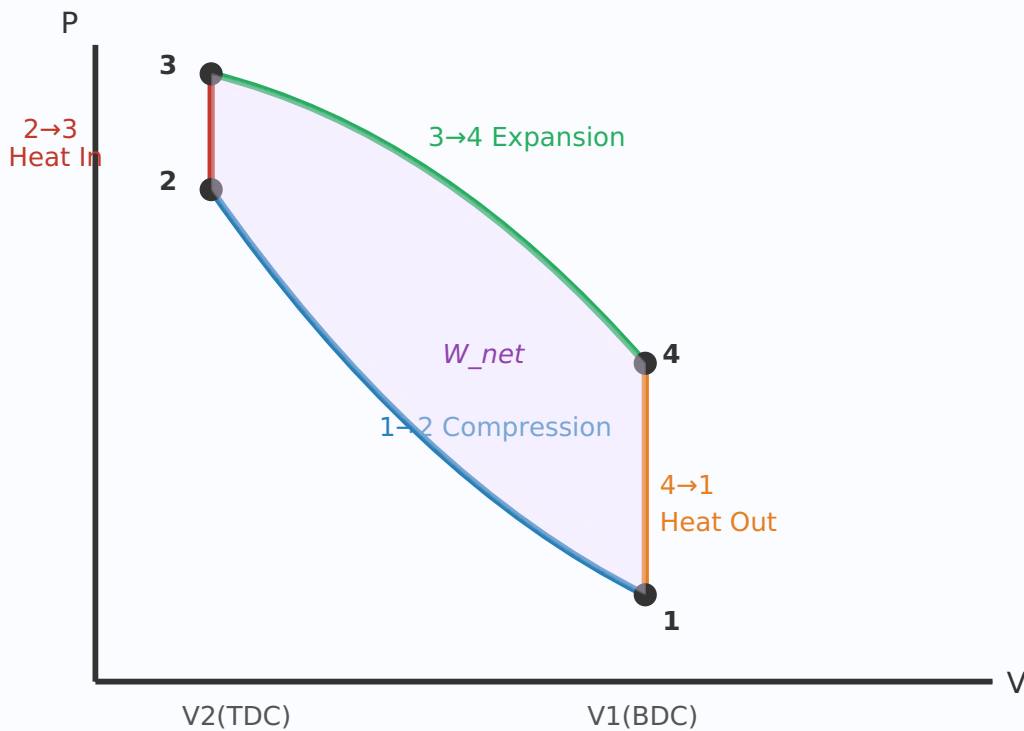
where r = Compression Ratio = V_1/V_2

γ = Ratio of specific heats = $C_p/C_v = 1.4$ for air

Work Output: $W_{\text{net}} = Q_{\text{in}} - Q_{\text{out}} = C_v(T_3 - T_2) - C_v(T_4 - T_1)$

Mean Effective Pressure (MEP): $p_m = W_{\text{net}} / (V_1 - V_2)$

FIG 10.1 - OTTO CYCLE P-V DIAGRAM



10.2 Diesel Cycle (Constant Pressure Cycle)

The **Diesel Cycle** is the ideal thermodynamic cycle for Compression Ignition (CI/diesel) engines. The key difference from the Otto cycle is that heat addition occurs at **constant pressure**.

- **Process 1→2**: Isentropic compression (higher compression ratio than Otto).
- **Process 2→3**: Constant Pressure heat addition (fuel injected; combustion).
- **Process 3→4**: Isentropic expansion (power stroke).
- **Process 4→1**: Constant Volume heat rejection.

Diesel Cycle - Efficiency Formula

$$\eta_{\text{Diesel}} = 1 - [1/(r^{\gamma}(\gamma-1))] \times [(rc^{\gamma} - 1) / (\gamma(rc - 1))]$$

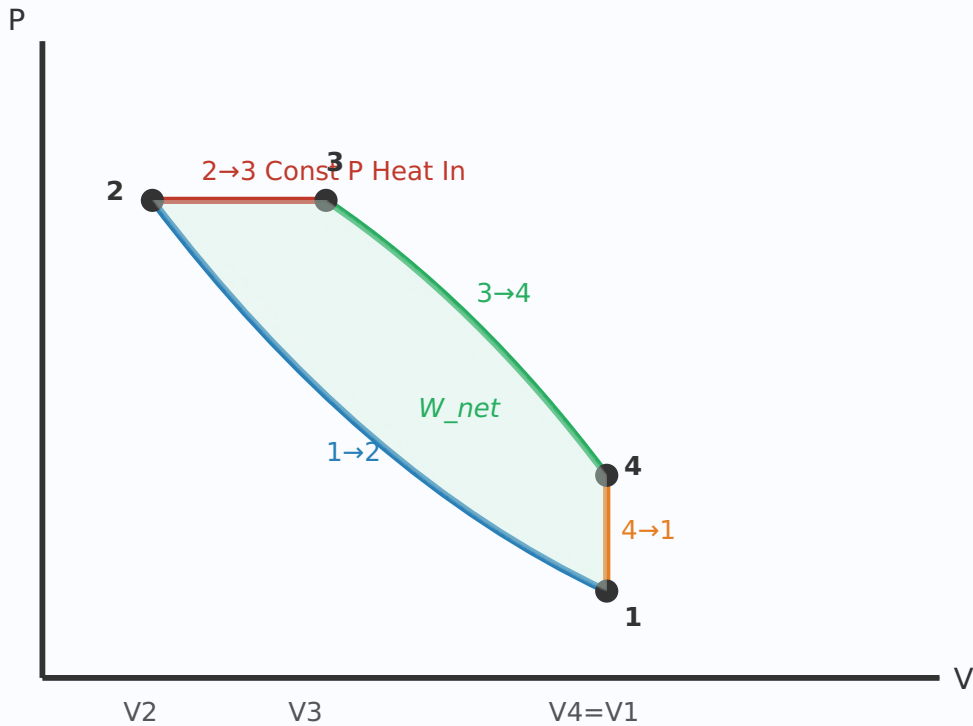
where r = Compression Ratio (V_1/V_2)

rc = Cut-off Ratio = V_3/V_2 (ratio of volume after to before combustion)

Note: For same compression ratio, $\eta_{\text{Otto}} > \eta_{\text{Diesel}}$

But Diesel engines use higher r (14-22) vs Otto (6-10), so practical Diesel η is higher

FIG 10.2 - DIESEL CYCLE P-V DIAGRAM



10.3 Otto vs Diesel Cycle Comparison

Parameter	Otto Cycle	Diesel Cycle
Fuel used	Petrol (Gasoline)	Diesel
Heat addition	Constant Volume	Constant Pressure
Compression Ratio	6 : 1 to 10 : 1	14 : 1 to 22 : 1
Ignition	Spark plug	Self-ignition (compression)
Efficiency (same r)	Higher	Lower
Practical efficiency	~25-30%	~35-45%
Application	Cars, bikes (SI engines)	Trucks, buses (CI engines)

Section 11: Refrigeration & Air Conditioning Cycle

Vapour Compression Refrigeration | COP | AC Principle

11.1 Refrigeration - Definition

Refrigeration is the process of maintaining a body or space at a temperature lower than its surroundings by continuously extracting heat from it. It works against the natural direction of heat flow (from cold to hot) by doing work.

11.2 Vapour Compression Refrigeration Cycle (VCRC)

The VCRC is the most widely used refrigeration cycle. It uses a **refrigerant** (Freon R-134a, R-22, R-410A) that alternately evaporates and condenses.

Four main components: Compressor, Condenser, Expansion Valve (Throttle), Evaporator.

11.3 Working of VCRC (Step-by-Step)

Step 1: Compressor (1→2): Low-pressure refrigerant vapour is compressed to high pressure and high temperature.

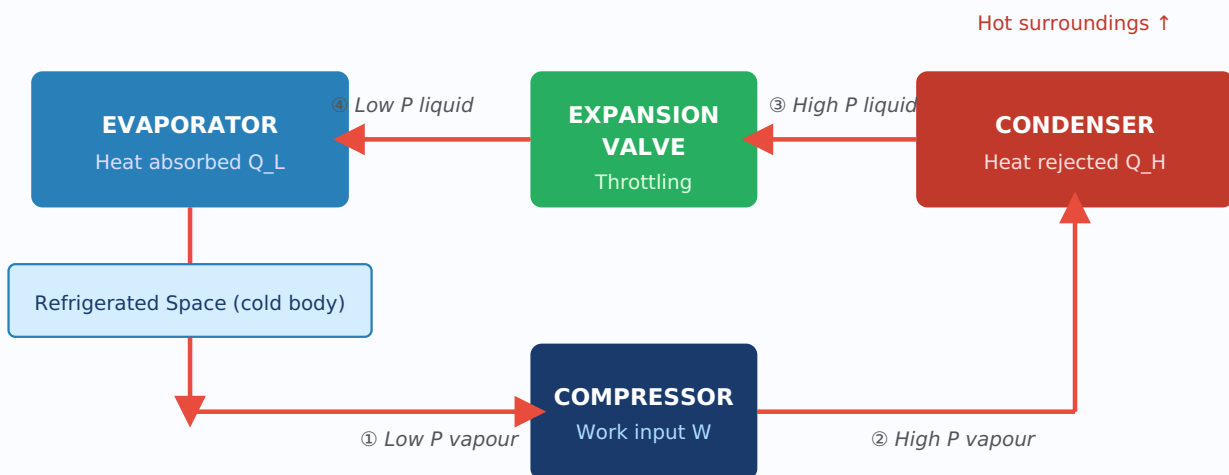
Step 2: Condenser (2→3): High-pressure hot vapour condenses to liquid by rejecting heat to surroundings (Q_H released).

Step 3: Expansion Valve (3→4): High-pressure liquid is throttled (expanded) to low pressure – temperature drops drastically.

Step 4: Evaporator (4→1): Low-pressure cold refrigerant absorbs heat from the refrigerated space (Q_L absorbed), evaporating into vapour.

Step 5: Cycle repeats – the refrigerated space is kept cold.

FIG 11.1 - VAPOUR COMPRESSION REFRIGERATION CYCLE DIAGRAM



Refrigeration - Key Formulae

$$\text{COP (Refrigerator)} = \frac{Q_L}{W} = \frac{Q_L}{(Q_H - Q_L)}$$

$$\text{COP (Heat Pump)} = \frac{Q_H}{W} = \frac{Q_H}{(Q_H - Q_L)}$$

$$\text{COP}_{\text{Carnot}} = \frac{T_L}{(T_H - T_L)} \quad (\text{ideal maximum})$$

$$\text{Refrigerating Effect} = m \times (h_1 - h_4) \text{ kJ/kg}$$

$$\text{Refrigerating Capacity in Tons} = \frac{\text{RE} \times m}{3.5} \text{ (kW)}$$

$$1 \text{ Tonne of Refrigeration} = 3.5 \text{ kW} = 210 \text{ kJ/min}$$

11.4 Air Conditioning (AC) Cycle

Air Conditioning controls temperature, humidity, air quality, and air circulation in a space for human comfort. A standard AC system uses the same VCRC with the addition of a **blower**, **air filter**, **thermostat**, and **duct network**.

- **Indoor Unit:** Contains the evaporator coil and blower fan – absorbs heat from room air.
- **Outdoor Unit:** Contains the compressor and condenser coil – rejects heat to outside.
- **Refrigerant Lines:** Connect indoor and outdoor units.
- **Thermostat:** Senses room temperature and controls compressor on/off.

Section 12: IC Engines - 2-Stroke & 4-Stroke

Internal Combustion Engines | Working Strokes | Comparison

12.1 IC Engine - Definition

An **Internal Combustion (IC) Engine** is a heat engine in which combustion of fuel occurs inside the engine cylinder, and the heat energy released is directly converted into mechanical work by the piston.

12.2 Main Parts of an IC Engine

- **Cylinder:** Cylindrical chamber where combustion takes place.
- **Piston:** Reciprocating member that transmits force to crankshaft.
- **Crankshaft:** Converts reciprocating motion to rotary motion.
- **Connecting Rod:** Links piston to crankshaft.
- **Valves:** Inlet and exhaust valves control gas flow.
- **Spark Plug:** (SI engine) Ignites air-fuel mixture.
- **Fuel Injector:** (CI engine) Injects fuel into cylinder.
- **Camshaft:** Controls valve timing.
- **Flywheel:** Stores and releases energy for smooth rotation.

12.3 Four-Stroke Engine - Working

Step 1: Suction Stroke: Piston moves from TDC to BDC. Inlet valve opens. Air-fuel mixture drawn into cylinder.

Step 2: Compression Stroke: All valves closed. Piston moves BDC to TDC. Mixture compressed (ratio 8:1 for petrol).

Step 3: Power (Expansion) Stroke: Near TDC, spark plug fires. Combustion pushes piston down to BDC. Work is produced.

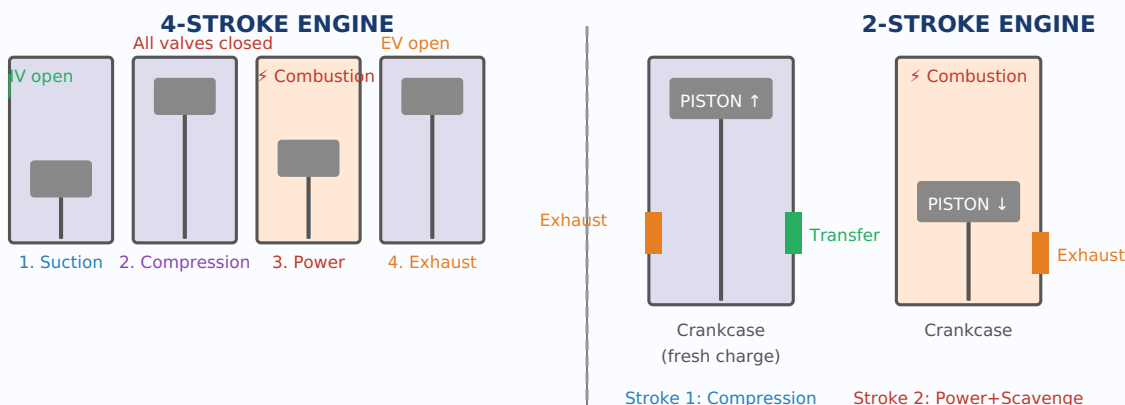
Step 4: Exhaust Stroke: Exhaust valve opens. Piston moves BDC to TDC, pushing out burnt gases. Cycle repeats.

12.4 Two-Stroke Engine - Working

Step 1: Compression & Transfer Stroke (Upward): Piston moves up. Air-fuel mixture compressed above piston. Below piston, fresh charge enters crankcase.

Step 2: Power & Exhaust Stroke (Downward): Combustion pushes piston down. Exhaust ports exposed – burnt gases escape. Transfer ports open – fresh charge enters cylinder (scavenging).

FIG 12.1 - 2-STROKE VS 4-STROKE ENGINE DIAGRAM (SCHEMATIC)



12.5 Comparison: 2-Stroke vs 4-Stroke Engine

Parameter	2-Stroke Engine	4-Stroke Engine
Power strokes	1 per revolution	1 per 2 revolutions
Power output	Higher for same size	Lower for same size
Fuel efficiency	Poor (fuel loss in scavenging)	Good
Lubrication	Oil mixed with petrol	Separate lubricating system
Valves	Ports (no valves)	Valves (inlet & exhaust)
Weight & size	Lighter and compact	Heavier and larger
Pollution	More (oil burning)	Less
Applications	Mopeds, chainsaws, outboard motors	Cars, trucks, generators
Thermal efficiency	Lower (~20-25%)	Higher (~30-40%)

Section 13: SI & CI Engines | Electric & Hybrid Vehicles

Spark Ignition vs Compression Ignition | EV Components | Hybrid Systems

13.1 SI Engine (Spark Ignition)

An **SI engine** uses a spark plug to ignite an air-petrol mixture. The fuel (petrol/gasoline) is mixed with air before compression. The Otto cycle is the ideal cycle for SI engines.

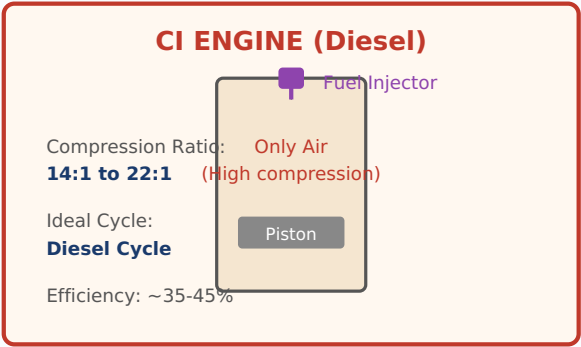
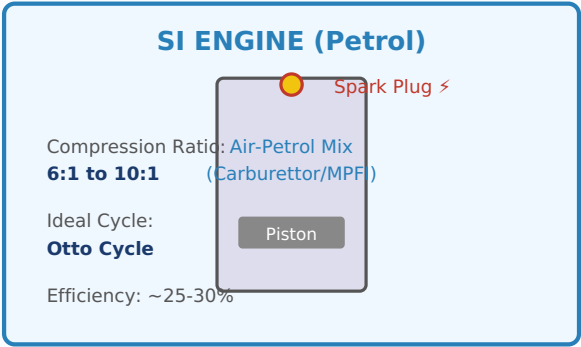
Key features: Low compression ratio (6-10:1), uses petrol, spark plug required, lighter and smoother operation, lower thermal efficiency (~25-30%).

13.2 CI Engine (Compression Ignition)

A **CI engine** uses the heat generated by high compression to ignite diesel fuel, which is injected near TDC. No spark plug is needed. The Diesel cycle is the ideal cycle for CI engines.

Key features: High compression ratio (14-22:1), uses diesel, fuel injector required, heavier and noisier, higher thermal efficiency (~35-45%).

FIG 13.1 - SI VS CI ENGINE COMPARISON DIAGRAM



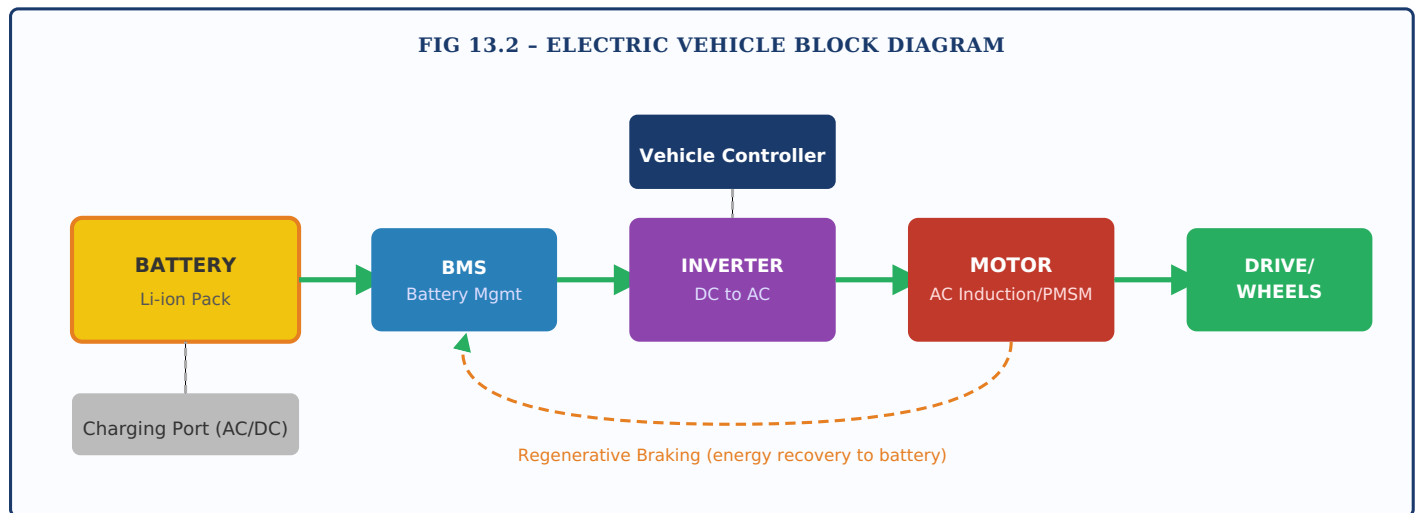
13.3 SI vs CI Engine Comparison Table

Parameter	SI Engine	CI Engine
Fuel	Petrol (Gasoline)	Diesel
Ignition	Spark plug (electrical)	Self-ignition (heat of compression)
Compression Ratio	6:1 - 10:1	14:1 - 22:1
Fuel introduction	With air (carburetor/MPFI)	Injected near TDC
Ideal Cycle	Otto Cycle	Diesel Cycle
Thermal Efficiency	25-30%	35-45%
Speed	High speed (2000-6000 rpm)	Lower speed (1000-3500 rpm)
Torque	Lower at low rpm	High torque at low rpm
Weight/Size	Lighter & compact	Heavier (higher pressures)
Noise & Vibration	Less noise	More noise and vibration
Applications	Cars, motorcycles, boats	Trucks, buses, ships, generators

13.4 Electric Vehicle (EV) Components

An **Electric Vehicle (EV)** is powered entirely by electric motors using energy stored in rechargeable battery packs. It

produces zero direct emissions.



13.5 Hybrid Vehicle Components

A **Hybrid Electric Vehicle (HEV)** combines an Internal Combustion Engine with an Electric Motor and Battery. The two power sources work together to optimise efficiency and reduce emissions.

- **ICE (Internal Combustion Engine):** Provides primary propulsion at high speeds; charges battery.
- **Electric Motor/Generator:** Assists ICE at acceleration; generates electricity during braking.
- **Battery Pack:** Smaller than full EV; stores recovered energy.
- **Power Split Device (PSD):** Planetary gear set that distributes power between ICE, motor, and wheels (Toyota HSD).
- **Power Control Unit (PCU):** Manages power flow between all components.
- **Regenerative Braking System:** Converts kinetic energy to electrical energy during braking.

Section 14: Questions & Answers

2-Mark | 5-Mark | 10-Mark | Exam-Oriented

14.1 Two-Mark Questions & Answers

Q1

2 Marks

Define Manufacturing Process and list its types.

Manufacturing is the process of converting raw materials into finished products using physical, chemical, or mechanical processes. The main types are: (1) Casting – pouring molten metal into moulds; (2) Forming – shaping by plastic deformation (forging, rolling, extrusion); (3) Joining – welding, brazing, soldering; (4) Machining – material removal by cutting (turning, drilling, milling); and (5) Advanced processes – CNC machining, 3D printing, and smart manufacturing.

Q2

2 Marks

What is COP of a refrigerator? Write the formula.

COP (Coefficient of Performance) of a refrigerator is the ratio of refrigerating effect (heat absorbed from cold body) to the net work input required. It is a measure of the effectiveness of a refrigeration system. Formula: $COP = Q_L / W = Q_L / (Q_H - Q_L)$, where Q_L = heat absorbed from cold body, Q_H = heat rejected to hot surroundings, W = work input. The Carnot COP = $T_L / (T_H - T_L)$.

Q3

2 Marks

State the differences between SI and CI engines.

SI (Spark Ignition) engines use petrol as fuel and require a spark plug to ignite the air-fuel mixture. They operate on the Otto cycle with a compression ratio of 6:1 to 10:1 and have efficiency of 25-30%. CI (Compression Ignition) engines use diesel fuel which self-ignites due to heat of compression. They operate on the Diesel cycle with compression ratio 14:1 to 22:1 and have higher efficiency of 35-45%. CI engines have more torque at low speeds and are used in heavy vehicles.

Q4

2 Marks

What is 3D Printing? Name its types.

3D Printing (Additive Manufacturing) is the process of creating three-dimensional solid objects from a digital CAD file by depositing material layer by layer. Unlike subtractive manufacturing, no material is removed; instead, material is added. Types include: (1) FDM – Fused Deposition Modelling, uses thermoplastic filament; (2) SLA – Stereolithography, uses UV-cured resin; (3) SLS – Selective Laser Sintering, uses powdered materials; and (4) DMLS – Direct Metal Laser Sintering, for metal parts.

Q5

2 Marks

Define thermal efficiency. Write the Otto cycle efficiency formula.

Thermal efficiency is the ratio of the net work output of a heat engine cycle to the heat supplied (Q_{in}). It measures how effectively an engine converts heat energy into work. $\eta_{thermal} = W_{net} / Q_{in} = 1 - (Q_{out} / Q_{in})$. For the Otto cycle specifically: $\eta_{Otto} = 1 - (1/r^{(\gamma-1)})$, where r = compression ratio (V_1/V_2) and γ = ratio of specific heats = $C_p/C_v = 1.4$ for air. Higher compression ratio gives higher efficiency.

14.2 Five-Mark Questions & Answers

Q6

5 Marks

Explain the Otto Cycle with a neat P-V diagram.

Otto Cycle is the ideal thermodynamic cycle for petrol (SI) engines, proposed by Nikolaus August Otto (1876). It consists of four reversible processes on a closed system of air (air standard cycle).

Process 1→2 (Isentropic Compression): The piston moves from BDC (Bottom Dead Centre) to TDC (Top Dead Centre). The air-fuel mixture is compressed without heat transfer (adiabatically). Temperature and pressure rise. $W = -\Delta U$.

Process 2→3 (Constant Volume Heat Addition): The spark plug fires near TDC. Combustion occurs rapidly at nearly constant volume. Heat Q_{in} is added. Pressure and temperature rise sharply. Volume remains constant (isochoric process).

Process 3→4 (Isentropic Expansion / Power Stroke): High-pressure gases expand adiabatically, pushing the piston from TDC to BDC. Work is done by the gases on the piston. Temperature and pressure fall.

Process 4→1 (Constant Volume Heat Rejection): At BDC, the exhaust valve opens. Heat Q_{out} is rejected at constant volume as burnt gases escape. Pressure falls to initial value. Cycle is complete.

Efficiency: $\eta = 1 - (1/r^{(\gamma-1)})$ where $r = V_1/V_2$ (compression ratio) and $\gamma = 1.4$ (for air). A higher compression ratio gives higher efficiency. Typical $r = 8$ to 10 for petrol engines, giving $\eta \approx 25\text{--}30\%$.

Important: The Otto cycle assumes air as the working medium (air standard analysis), perfect combustion, and no friction losses.

Q7

5 Marks

Explain the Casting Process with a neat sketch.

Casting is a primary manufacturing process in which molten metal is poured into a pre-shaped mould cavity and allowed to solidify to form the desired shape.

Types: Sand casting, Die casting, Investment casting, Centrifugal casting. Sand casting is the most common.

Pattern: A replica of the final product made from wood, aluminium, or plastic is used to create the mould impression.

Mould: Consists of two halves – the COPE (upper half) and DRAG (lower half). The parting line is the interface between the two halves.

Gating System: Consists of Sprue (vertical channel), Runner (horizontal channel), and Gate (entry to cavity) through which molten metal flows.

Riser: A reservoir of molten metal that feeds the cavity during solidification to compensate for shrinkage.

Process Steps: (1) Pattern making → (2) Mould preparation → (3) Melting metal → (4) Pouring → (5) Solidification → (6) Shakeout (breaking mould) → (7) Fettling (cleaning).

Applications: Engine blocks, cylinder heads, pump casings, machine tool bases, artistic objects.

Q8

5 Marks

Compare 2-Stroke and 4-Stroke engines with neat diagrams.

Both 2-stroke and 4-stroke engines are IC engines that convert chemical energy of fuel into mechanical work.

4-Stroke Engine: Completes one thermodynamic cycle (power stroke) every two revolutions of the crankshaft (four strokes). Strokes: Suction → Compression → Power → Exhaust. Has valves for gas exchange. Better fuel efficiency, less pollution. Used in cars, trucks, generators.

2-Stroke Engine: Completes one thermodynamic cycle every single revolution of the crankshaft (two strokes). Uses ports instead of valves. Scavenging (simultaneous exhaust and intake) occurs at BDC. Power-to-weight ratio is higher but fuel efficiency is lower due to scavenging losses.

Key Differences: Power frequency (2-stroke gives power every revolution vs every 2 revolutions in 4-stroke). Lubrication (oil mixed with fuel in 2-stroke). Mechanical simplicity (2-stroke has no valves). Fuel efficiency (4-stroke is better). Emissions (2-stroke produces more smoke).

14.3 Ten-Mark Questions & Answers

Q9

10 Marks

Explain the Vapour Compression Refrigeration Cycle with a neat diagram. Also calculate COP if heat absorbed is 150 kJ and work input is 50 kJ.

Introduction: The Vapour Compression Refrigeration Cycle (VCRC) is the most widely used cycle for refrigeration and air conditioning systems. It uses a refrigerant (R-134a, R-410A) which alternately evaporates and condenses to absorb and reject heat.

Components and Functions:

- 1. Compressor:** The heart of the system. Draws in low-pressure refrigerant vapour from the evaporator and compresses it to high pressure and high temperature. Work input W is applied here (via electric motor). Process: 1→2 (isentropic compression).
- 2. Condenser:** High-pressure, high-temperature refrigerant vapour enters the condenser. Heat Q_H is rejected to the surroundings (ambient air or water). Refrigerant condenses from vapour to liquid state. Process: 2→3 (constant pressure heat rejection).
- 3. Expansion Valve (Throttle):** High-pressure liquid refrigerant is throttled (expanded suddenly) through a narrow orifice. This causes a drastic drop in pressure and temperature (refrigerant partially evaporates). Process: 3→4 (isenthalpic throttling).
- 4. Evaporator:** Low-pressure, low-temperature refrigerant absorbs heat Q_L from the refrigerated space (cold body). Refrigerant completely evaporates. The refrigerated space is cooled. Process: 4→1 (constant pressure heat absorption).

Energy Balance: $Q_H = Q_L + W$ (by first law of thermodynamics).

Formulae: $\text{COP}_{\text{refrigerator}} = Q_L / W = Q_L / (Q_H - Q_L)$. $\text{COP}_{\text{Carnot}} = T_L / (T_H - T_L)$. 1 Ton of Refrigeration = 3.5 kW = 210 kJ/min.

Numerical Solution: Given $Q_L = 150$ kJ, $W = 50$ kJ. $\text{COP} = Q_L / W = 150 / 50 = 3.0$. This means for every 1 kJ of work input, 3 kJ of heat is removed from the cold space. $Q_H = Q_L + W = 150 + 50 = 200$ kJ.

Applications: Household refrigerators, air conditioners, cold storage warehouses, ice plants, cryogenic cooling, food preservation.

Q10

10 Marks

Explain Smart Manufacturing. What are its key technologies and benefits?

Introduction: Smart Manufacturing (Industry 4.0) is the fourth industrial revolution, characterised by the integration of digital, physical, and biological systems in manufacturing. It leverages advanced technologies to create self-aware, self-learning, and self-adaptive manufacturing processes.

- 1. Internet of Things (IoT):** Sensors embedded in machines collect real-time data on temperature, vibration, speed, and quality. This data is transmitted over networks for monitoring and decision-making. Predictive maintenance reduces unplanned downtime.
- 2. Artificial Intelligence and Machine Learning:** AI algorithms analyse vast production data to optimise processes, detect defects (computer vision), forecast demand, and schedule production dynamically.
- 3. Digital Twin Technology:** A virtual replica (digital twin) of the physical manufacturing system is created. Engineers simulate, test, and optimise processes virtually before implementation, reducing errors and cost.
- 4. Robotics and Collaborative Robots (Cobots):** Programmable robots handle repetitive, dangerous, or precision tasks. Cobots work safely alongside human workers, increasing flexibility and productivity.
- 5. Cloud Computing and Big Data:** Manufacturing data is stored and processed in the cloud, enabling remote monitoring, analytics, and collaboration across global supply chains.
- 6. Additive Manufacturing (3D Printing):** Integrated with digital workflows for on-demand, customised production of complex parts without traditional tooling.

7. Augmented Reality (AR): AR glasses and devices guide workers in assembly, maintenance, and quality inspection with real-time overlaid instructions.

Benefits: Increased productivity and efficiency. Reduced material waste and energy consumption. Improved product quality and consistency. Enhanced worker safety. Greater flexibility and mass customisation. Predictive (not reactive) maintenance. Faster time-to-market for new products.

Applications: Tesla's Gigafactory (automated EV assembly), Siemens' smart factory (digital twins), pharmaceutical industry (automated quality control), semiconductor fabs, aerospace component manufacturing.

Section 15: Quick Revision Sheet & References

Key Formulae | Important Diagrams | One-Page Summary

15.1 Key Formulae at a Glance

🏭 Manufacturing	🔥 Thermal & Cycles	❄️ Refrigeration	⚙️ IC Engines
- Solidification: $t = B(V/A)^2$ - Cutting Speed: $V = \pi DN/1000$ - MRR = $f \times d \times V$ - True Strain: $\epsilon = \ln(L_f/L_0)$ - Weld Heat: $H = VI \times 60 / (S \times 1000)$	$\eta_{Otto} = 1 - 1/r^{(\gamma-1)}$ $\eta_{Diesel} = 1 - [1/r^{(\gamma-1)}] \times [(rc^{\gamma-1}) / (\gamma(rc-1))]$ $\eta_{Carnot} = 1 - T_L/T_H$ 1st Law: $Q = \Delta U + W$ - Boiler $\eta = ms(hs-hfw)/(mf \times CV)$	$COP_{ref} = Q_L/W = Q_L/(Q_H-Q_L)$ $COP_{HP} = Q_H/W$ $COP_{Carnot} = T_L/(T_H-T_L)$ 1 TR = 3.5 kW = 210 kJ/min $Q_H = Q_L + W$	- Compression ratio $r = V_{BDC}/V_{TDC}$ $MEP = W_{net} / (V_1-V_2)$ - Power: $P = T \times \omega$ (kW) 4-stroke: 1 power/2 rev 2-stroke: 1 power/1 rev

15.2 Process Steps Summary

Process	Steps Summary	Key Product
Sand Casting	Pattern → Mould → Pour → Solidify → Remove → Fettling	Engine block
Forging	Heat billet → Place in die → Apply force → Remove → Trim	Crankshaft
Arc Welding	Strike arc → Form weld pool → Cool → Chip slag	Steel structures
Lathe Turning	Chuck workpiece → Set rpm → Feed tool → Measure	Shafts, pins
FDM 3D Print	CAD → STL → Slice → Heat filament → Deposit layers	Prototypes
VCRC (Refrig.)	Compress → Condense → Expand → Evaporate → Repeat	Cold space
4-Stroke Engine	Suction → Compression → Power → Exhaust	Mechanical work
Otto Cycle	1→2 Compress → 2→3 Heat add (const.V) → 3→4 Expand → 4→1 Heat reject	SI engine work

15.3 Important Diagram List

#	Diagram Name	Key Labels to Remember
1	Sand Casting	Cope, Drag, Sprue, Gate, Riser, Mould cavity, Parting line
2	Arc Welding	Electrode, Arc, Weld pool, Slag, Base metal, Power supply
3	Lathe Machine	Headstock, Tailstock, Carriage, Chuck, Tool post, Bed, Feed rod
4	CNC System	Input → MCU → Drive → Machine Tool → Output; Feedback loop
5	FDM 3D Printer	Filament spool, Print head, Heater, Nozzle, Build plate, Layers
6	Cochran Boiler	Furnace, Fire tubes, Steam space, Smoke box, Safety valve
7	Otto Cycle PV	Points 1,2,3,4; Constant volume lines; W_{net} area
8	Diesel Cycle PV	Points 1,2,3,4; Constant pressure line 2→3; Cut-off ratio rc
9	VCRC	Compressor, Condenser, Expansion valve, Evaporator; Q_H, Q_L, W
10	EV Block Diagram	Battery → BMS → Inverter → Motor → Wheels; Regen braking

15.4 Differences at a Glance

Comparison	Option A	Option B	Key Difference

2-Stroke vs 4-Stroke	1 power/revolution	1 power/2 revolutions	Frequency of power
SI vs CI Engine	Petrol + Spark plug	Diesel + Self-ignition	Fuel & ignition
Otto vs Diesel	Const. Volume heat add	Const. Pressure heat add	Mode of combustion
Fire tube vs Water tube	Gases in tubes	Water in tubes	What flows in tubes
Casting vs Forging	Liquid → Mould	Solid → Deformation	State of material
EV vs Hybrid	Only battery/motor	ICE + Motor + Battery	Power source

15.5 Textbook References

Main Textbooks:

1. V. Ganesan – *Internal Combustion Engines*, 4th Edition, Tata McGraw-Hill Education.
2. S.S. Rattan – *Theory of Machines*, S. Chand Publications, Tata McGraw-Hill.
3. Jonathan Wicker & Kemper Lewis – *Introduction to Mechanical Engineering*, Cengage Learning.

Reference Books:

4. G. Shanmugam & M.S. Palanisamy – *Basic Civil & Mechanical Engineering*, Anuradha Publications.
5. Mahesh M. Rathore – *Thermal Engineering*, Tata McGraw-Hill Education.
6. L. Jyothish Kumar & Pulak Pandey – *3D Printing & Additive Manufacturing*, Cambridge University Press.
7. Appu Kuttan K.K – *Robotics*, I.K. International Publishing House Pvt. Ltd.

Best of Luck for Your Examinations!

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Study Hard · Understand Concepts · Draw Neat Diagrams · Write Structured Answers